Noetic Diffusion Theory Compendium

A unified mathematical formalization of the NDT article family

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Abstract This compendium consolidates the mathematical backbone of the Noetic Diffusion Theory article family into one notation. It distinguishes the controlled transition law from its non-equivalent local-dynamics families: drift, diffusion geometry, chart Jacobians, finite-time response, stochastic short-horizon predictive spread, destination probabilities, attractor structure, and finite-amplitude resilience. It also records the base and stratified Meta–Noetic Phase Space (MNPS), the NeuralManifoldDynamics measurement contract, and the speculative micro-to-meso accessibility bridge. The goal is not to reproduce each paper verbatim, but to provide one dependency-ordered reference in which latent theory, measurement charts, local estimators, and validity boundaries remain distinct [[1]; [2]; [3]; [4]].

1. Global Notation and Layering

We fix one canonical notation throughout. The latent noetic process evolves on a manifold $\mathcal{M}$, while empirical analyses work in local coordinates derived from EEG, iEEG, or fMRI measurements. This opening section therefore serves as the common dictionary for the rest of the compendium: first the latent object, then the chart, then the empirical observation layer, and only after that the local dynamical operators that act on the charted state.

$$X_t \in \mathcal{M}, \quad x_t = [m_t, d_t, e_t]^T \in \mathbb{R}^3 \tag{1}$$

$$x_t^{(9)} = [m_a, m_e, m_o, d_n, d_l, d_s, e_e, e_s, e_m]^T \in \mathbb{R}^9 \tag{2}$$

The observational layer is separated from the latent dynamics:

$$Y_t = g(X_t) + \varepsilon_t \tag{3}$$

Equations 1 and 2 define the canonical and stratified chart coordinates, while 3 places both charts below the measurement map. Here Y_t denotes measured neural data, g is a measurement map, and ε_t is observation noise. Theoretical dynamics live in X_t; MNPS and Stratified MNPS are empirical charts constructed from short windows of Y_t.

To minimize cross-paper ambiguity, the canonical axes are:

- m: metastability / mobility / rhythmic coordination.
- d: deviation from an optimal integration–segregation balance.
- e: entropy / entropic energy / complexity pressure.

Legacy triplets such as (E, R, D) are treated as manuscript-local precursors that are monotonically aligned to (m, d, e) at the compendium level. This is not merely cosmetic. A major source of friction across the article family is that the same conceptual roles are sometimes named differently depending on whether a manuscript is emphasizing latent theory, empirical measurement, or a transitional draft vocabulary.

We also separate five layers that are sometimes blended in the source papers:

- **Latent theory**: stochastic dynamics on $\mathcal{M}$.
- **Microstructural substrate**: branch/spine access, slow structural accessibility, and eligibility-like consolidation variables.
- **Meso export**: lower-dimensional accessibility charts exported from branch-resolved microstates.
- **Measurement chart**: exported 3D or 9D coordinates used in analysis.

- **Operational contract**: release-fixed projections, derivatives, local dynamics, dynamical-family exports, and validity constraints used by `NeuralManifoldDynamics`.

2. Core NDT Dynamics

2.1. Controlled diffusion as a modelling class

The working process class of current NDT is a controlled diffusion, not a pure gradient flow and not a fitted neural stochastic differential equation. Let c_t denote a contextual or control state:

$$dX_t = b(X_t, t \mid c_t)dt + \Sigma(X_t, t \mid c_t)dW_t \tag{4}$$

Here b is local deterministic drift, Σ determines the amplitude and orientation of stochastic forcing, and W_t is Brownian motion. This is an effective continuous-state approximation represented by 4. It does not assert that neural fluctuations are literally Brownian or that all relevant dynamics are Markovian. Hybrid, coloured-noise, delayed, and non-Markov formulations remain compatible when required by the observation scale.

2.2. Transition-law family

The central object is the control-dependent transition law, not a single accessibility scalar. The generator of 4, acting on a test function f, is

$$\mathcal{L}_{t \mid c} f(\xi) = b(\xi, t \mid c) \cdot \nabla f(\xi) + \frac{1}{2} \operatorname{tr}\big(a(\xi, t \mid c)\nabla^2 f(\xi)\big) \tag{5}$$

with

$$a(\xi, t \mid c) = \Sigma(\xi, t \mid c)\Sigma(\xi, t \mid c)^\top \tag{6}$$

The diffusion tensor in 6 is the covariance-rate term appearing in the generator 5. The dummy variable ξ is a realization of the latent state X_t. For a control or history protocol γ over a finite interval, define

$$K^{X \mid \gamma}_{t:t+h}(\xi, A) = P\Big(X_{t+h} \in A \mid X_t = \xi \text{ and } \gamma_{[t:t+h]}\Big) \tag{7}$$

Empirical work generally estimates the chart-conditional counterpart

$$K^{x \mid \gamma}_{t:t+h}(x, B) = P\Big(x_{t+h} \in B \mid x_t = x \text{ and } \gamma_{[t:t+h]}\Big) \tag{8}$$

The latent law 7 and chart law 8 are therefore separate objects linked by the measurement/chart contract. When this compendium writes K^h without a superscript, it means the chart-conditional law K^x unless a latent argument is explicit. K^h is the law of the state at horizon h, not the full path-space measure. The family $\{K^h\}_{h>0}$ is the primary finite-horizon object. A snapshot-conditioned kernel with fixed c_t is only a special case; history may instead require an augmented state or protocol-path conditioning.

The named local-dynamics families answer different questions:

- b is the infinitesimal conditional mean rate.
- a is the infinitesimal conditional covariance rate.
- J is the local derivative of drift in a declared chart.
- Finite-time response follows a local Jacobian field over a stated horizon.
- Stochastic short-horizon reachability summarizes finite-horizon predictive spread.
- The committor q is a destination or hitting probability.
- Attractors describe long-time organization.
- FAR describes response to a finite displacement.

These are different kinds of object, not interchangeable views of one endpoint distribution. Rhythm, thalamus, history, body, and metabolism are candidate changes to $\mathcal{L}$ or K, not additional definitions of NDT.

2.3. Three core commitments

The current theory separates three commitments that should not be merged:

- **NDT-C1 (reduced-state insufficiency):** current location in a reduced measurement chart x_t need not determine the latent future law K^X.
- **NDT-C2 (non-redundant accessibility):** in at least some neural regimes, functionals of the chart-conditional law provide reproducible out-of-sample information not contained in current chart location or standard static and spectral summaries.
- **NDT-C3 (conditional control):** context, history, or physiology can alter $K^{x \mid \gamma}_h(\cdot \mid x)$ among observations matched on current chart state.

C1 is a sufficiency claim about a reduced chart and is not distinctive by itself. The empirical content is C2 and C3: whether future-law quantities add reproducible information, and whether predefined controls systematically reshape those futures at matched measured state. C3 is not the generic fact that adding any covariate can improve prediction.

2.4. Drift split and diffusion geometry

The diffusion tensor is not restricted to a scalar noise amplitude. Its orientation, effective dimensionality, anisotropy, and magnitude can vary independently of the local mean drift. A structured potential component may be written as 9:

$$b(\xi, t \mid c) = -M(\xi, t \mid c)\nabla V(\xi, t \mid c) + u_{\mathrm{nc}(\xi, t \mid c)} \tag{9}$$

Here $M \succ 0$ is a mobility tensor and V is an NDT potential. The residual u_{nc} permits circulation-like or asymmetric flow. Equation 9 changes the drift term b in 4; it does not replace the diffusion tensor 6. This decomposition is not uniquely identified from trajectories, and NDT does not impose a fluctuation–dissipation relation between M and a. A state-dependent mobility can itself produce non-symmetric chart Jacobians; asymmetry is not sufficient evidence for a non-conservative mechanism.

2.5. Rhythmic variance control as a special case

Rhythmic Variance Control (RVC) is the minimal isotropic accessibility-control case. For an operational rhythmic proxy $r(t) \in [0, 1]$,

$$\sigma(t) = \sigma_{\min} + \sigma_0(1 - r(t)) \tag{10}$$

High $r(t)$ reduces isotropic stochastic amplitude in this special case, while lower $r(t)$ permits broader exploration. The general control hypothesis is broader:

$$r(t), c(t) \to \{b_t, \Sigma_t, M_t, \text{transition gates}\} \tag{11}$$

Equation 10 is the scalar-variance limit of the broader control map 11. RVC therefore occupies one entry in the control layer; it is not the definition of NDT and does not establish causal denoising.

2.6. Formalization boundary

Regularity assumptions are local and contract-specific. Smoothness, finite moments, support, time-step, and observation-noise conditions must be declared before a generator or estimator is treated as well posed. The compendium provides notation for those objects; it does not claim that every theoretical quantity has a validated empirical estimator.

3. Measurement Layer: Base MNPS and Stratified MNPS

3.1. Base 3D Chart

The base measurement chart is the 3D MNPS coordinate vector in 1. The stratified vector introduced in 2 resolves subcoordinate masking, while 3 makes explicit that both are measurement-layer objects. The source papers consistently motivate the base chart as the minimal low-dimensional summary of noetic state, while also warning that composite 3D coordinates can mask compensatory shifts among latent mechanisms [[1]; [2]]. The practical payoff of starting with 3D is interpretability; the practical cost is that opposite subcoordinate movements can cancel one another and leave the composite apparently unchanged.

3.2. Canonical 9D Stratified Chart

To resolve that masking, the stratified chart expands each axis into three sub-coordinates. This is the key move that turns MNPS from a coarse state map into a mechanism-sensitive chart: the question shifts from "where is the system in (m, d, e)?" to "which sub-processes are pushing that position?" The vector in 2 is unpacked into the nine operational families below. [[2]].

- m family: m_a, m_e, m_o.
- d family: d_n, d_l, d_s.
- e family: e_e, e_s, e_m.

The principal definitions collected from the Stratified MNPS manuscript are:

$$m_a = \mathrm{SR} * \mathrm{alpha_attn_mean} \tag{12}$$

$$m_e = \mathrm{SR} * \left(\max\left(z_{L(w)}, 0\right)\right)_W \tag{13}$$

$$\mathrm{SC}(w) = \frac{\sum_f f P(f,w)}{\sum_f P(f,w)}, \quad m_o = \frac{\mathrm{std}_{W(\mathrm{SC})}}{\mathrm{avg}_{W(\mathrm{SC})}} \tag{14}$$

$$H(t_h, w) = e^{-t_h L(w)}, \quad d_n = 1 - \mathrm{norm}\left(\mathrm{avg}_{W\left(\frac{\mathrm{Tr}(H(t_h,w))}{|V|}\right)}\right) \tag{15}$$

$$d_l = \mathrm{avg}_{W\left(C_{i(w)}\right)} \tag{16}$$

$$d_s = \mathrm{std}_{u(\mathrm{proj}(b_u))} \quad \text{or} \quad d_s = \mathrm{avg}_{W(1-\rho(\mathrm{RDM}(s),\mathrm{RDM}(b(w))))} \tag{17}$$

$$e_e = \frac{1}{S}\sum_{s=1}^{S} \mathrm{SampEn}_s \tag{18}$$

$$e_s = \mathrm{H}_{\mathrm{perm}(\varphi, m_p, \tau)} \tag{19}$$

$$e_m = z\left(\frac{\mathrm{P}_{\mathrm{beta}} + \mathrm{P}_{\mathrm{gamma}}}{\mathrm{P}_{\mathrm{alpha}} + \mathrm{P}_{\mathrm{theta}}}\right) \quad \text{or} \quad e_m = z\Big(\mathrm{avg}_{W\left(E_{g(w)}\right)}\Big) \tag{20}$$

Interpretively:

- m_a tracks attentional state-switch mobility.
- m_e tracks affective or limbic mobility.
- m_o tracks oscillatory flexibility.
- d_n tracks network-level diffusivity via graph heat propagation.
- d_l tracks local coupling or clustering.
- d_s tracks representational spread or dispersion.
- e_e tracks multiscale entropy.
- e_s tracks phase-order entropy.
- e_m tracks an efficiency or energetic proxy.

Several of these objects are explicitly tied to established signal-processing families in the source paper: multiscale sample entropy for e_e [[5]], graph connectivity measures such as wPLI or related phase locking summaries for network-sensitive coordinates [[6]; [7]], and spectral-centroid variability for oscillatory flexibility [[2]].

3.3. Recomposition to 3D

The theoretical default is a robust recomposition. This is where the stratified manuscript deliberately departs from naive averaging: if the 9D chart exists to expose masked redistributions, recomposition back to 3D should not reintroduce avoidable sensitivity to outliers or unstable subaxes [[2]].

$$m = \mathrm{robust_mean}(m_a, m_e, m_o), \quad d = \mathrm{robust_mean}(d_n, d_l, d_s), \quad e = \mathrm{robust_mean}(e_e, e_s, e_m) \tag{21}$$

with a weighted alternative

$$m = w_a m_a + w_e m_e + w_o m_o, \quad d = w_n d_n + w_l d_l + w_s d_s, \quad e = w_{\mathrm{ee}} e_e + w_{\mathrm{es}} e_s + w_{\mathrm{em}} e_m \tag{22}$$

under nonnegative weights summing to one within each family.

Equations 21 and 22 make the recomposition rule explicit: robustness is the default, while weighted recomposition is an auditable alternative. Neither rule changes the fact that the underlying measurements originate in the stratified chart.

The common statistical discipline across the papers is:

- compute inferences on within-subject robust z-values,
- reserve $[0, 1]$ scaling for reporting and visualization,
- treat monotone transforms as preserving order but not absolute scale.

3.4. Mapping from Latent Theory to Measurement

The stratified article and Churchland appendix propose schematic monotone bridges from latent NDT quantities to measured coordinates. These are legacy conceptual bridges, not identified NDT 1.0 estimators. They are useful only when they prevent the measurement layer from floating free of the theory: they make explicit how empirical feature families are supposed to stand in for latent NDT quantities without claiming one-to-one identity [[2]; [8]; [9]].

$$e_{e(X_t)} \approx h_{e(E(X_t))}, \quad e_{s(X_t)} \approx h_{s(r(X_t))}, \quad d_{n(X_t)} \approx \mathrm{norm}\Big(\mathrm{Tr}\Big(G_{t(X_t)}\Big)\Big) \tag{23}$$

and, more qualitatively,

$$d_{l(X_t)} \approx h_{l(\kappa(X_t))}, \quad m_{o(t)} \approx \mathrm{Var}_{W(\mathrm{SC}(\sigma(t)))}, \quad m_{a(t)} \approx h_{a(\text{precision pulses aligned to theta})} \tag{24}$$

These bridge functions are not themselves directly identified from data; they formalize a monotone link between latent dynamics and measurement-layer coordinates.

3.5. NeuralManifoldDynamics as a measurement contract

NeuralManifoldDynamics (NMD) implements selected MNPS and local-dynamics quantities through versioned preprocessing, feature construction, chart projection, and estimator contracts. NMD is an operational measurement layer, not NDT itself. The theory specifies latent X_t, the law $\mathcal{L}$, the transition kernels, and the questions asked of them; NMD specifies particular observation charts, serialized paths, estimators, and validity diagnostics [[4]].

The path and schema inventory below is an implementation snapshot and may be ahead of the cited contract manuscript. In the nomenclature used here, `mndm.dynamics` denotes Jacobian-derived local-dynamics surfaces, whereas `mndm.dynamical_families` denotes the separate diffusion, destination, and resilience families. Neither namespace identifies a latent biological process.

$$\mathcal{X}_t = [m_t, d_t, e_t], \quad \mathcal{X}_t^{(9)} = [m_a, m_e, m_o, d_n, d_l, d_s, e_e, e_s, e_m] \tag{25}$$

Equation 25 restates the chart families at the measurement-contract boundary. The current release treats 3D coordinates as a deterministic weighted projection of the 9D chart:

$$x_t = P_{\text{fixed}} x_t^{(9)} \tag{26}$$

with one active EEG-style policy instance summarized as

$$m = 0.62 m_a + 0.55 m_e + 0.45 m_o \tag{27}$$

$$d = 0.50 d_n + 0.82 d_l + 0.28 d_s \tag{28}$$

$$e = 0.85 e_e + 0.62 e_s + 0.03 e_m \tag{29}$$

Equations 27, 28, and 29 instantiate 26 for one policy. These coefficients are best read as one explicit policy example, not as universal modality-independent constants. In the implemented pipeline, `mnps_3d` is produced by a deterministic config-resolved projection from `coords_9d`: runtime configuration supplies a `v1_mapping`, feature-level weights are mapped onto available 9D subcoordinates, and the projection is renormalized over finite support when some subcoordinates are missing. The result is still a fixed auditable linear contract within a run, but its exact matrix may vary across modality or dataset policies [[4]].

The operational contract additionally fixes a serialized derivative `mnps_3d_dot`, coverage-aware renormalization when subcoordinates are missing, validity thresholds for local estimation, and explicit separation between `mnps_3d` and `coords_9d`. The derivative is a chart-level estimator; it is not automatically the latent drift b in 4.

This release-fixed projection is not claimed to be the unique true ontology of the manifold; it is an auditable measurement contract chosen to reduce silent chart drift across analyses. In that sense the compendium distinguishes clearly between a mathematical chart and a versioned export contract: the former is a theoretical coordinate system, the latter is a reproducibility device.

The public reference implementation for this measurement layer is `https://github.com/NoeticDiffusion/NeuralManifoldDynamics`. At the compendium level this matters because the object being standardized is not merely a set of names but a release-fixed ingest contract: feature normalization, coordinate export, derivative serialization, local neighborhoods, local-dynamics families, and validity diagnostics must be auditable before downstream claims can be compared across datasets.

3.6. Local-dynamics families in the contract

The local-dynamics surface is organized into two related groups. The first group contains Jacobian-derived families built from the chart flow or from one-step transition residuals:

- `mndm.jacobian_metrics.v1` stores chart-level metrics derived from `J_hat`, including spectral abscissa α, numerical abscissa ω, reactivity gap $\Delta_{\text{react}} = \omega - \alpha$, symmetric-rate summaries, rotation, and stable-reactive flags.

- `mndm.finite_time_response.v1` propagates a declared local Jacobian field over finite horizons. Its peak-gain analogue is `g_peak_over_horizons`; it is a model-derived tangent-map summary, not a perturbational FAR measurement or a held-out predictor.
- `mndm.residual_covariance_proxy.v1` records a residual covariance proxy. Derivative-residual covariance is not the diffusion tensor a and is not an admissible W_Q without an explicit conversion contract.
- `mndm.stochastic_reachability.v1` is reserved for covariance propagation driven by admissible one-step transition innovation W_Q. No silent derivative-residual-to-Q conversion is permitted, so this surface remains gated rather than implied by the presence of `J_hat`.

The second group is the separate `mndm.dynamical_families` surface:

- `mndm.diffusion_geometry.v1` estimates a chart-dependent increment-covariance object. It does not become an Itô tensor merely because it is called diffusion, and it does not establish drift–diffusion alignment when an independently qualified drift is absent.
- `mndm.committor.v1` represents destination or hitting-probability analyses. It requires declared competing regimes, disjoint support, sufficient independent segments, and an estimator qualification; stage labels alone are not committor truth.
- `mndm.finite_amplitude_resilience.v1` represents observed response to a finite perturbation. It requires a declared perturbation protocol and cannot be inferred from Jacobian, finite-time-response, or spontaneous-trajectory summaries.

The Jacobian surface is the operational counterpart of the chart derivative defined in 45 and estimated locally as 53. The diffusion and destination families instead refer to the covariance and hitting-probability objects in 6 and 70; they are not derived automatically from a Jacobian.

The two groups must not be collapsed into one chain. In particular, FTR is not FAR, a Jacobian residual is not a, and an NMD reachability export is not the same object as a study-specific predictive-spread analysis.

3.7. Implemented HDF5 surfaces

The canonical H5 write path serializes a measurement-layer contract rather than the full theoretical stack. This matters because many downstream discussions implicitly assume that every mathematically named object in the article family already exists as a standard serialized dataset. The ingest-side revision makes clear that this is not yet true: the H5 contract is narrower than the full compendium, and that narrowing should be stated explicitly [[4]].

On the main path the following objects are written as first-class datasets or groups:

- root series `time`, `mnps_3d`, and `mnps_3d_dot`,
- feature surfaces `features_raw` and `features_robust_z`,
- optional alignment/support datasets such as `z`, `window_start`, `window_end`, `events/*`, and `nn/indices`,
- labels under `labels/stage` and additional `labels/*`,
- primary dynamics under `jacobian/J_hat`, `jacobian/J_dot`, `jacobian/centers`, and optional `jacobian/diagnostics/*`,
- stratified outputs under `coords_9d/values`, `coords_9d/names`, and optional `jacobian_9D/*`,
- Jacobian-derived surfaces under `jacobian/derived_metrics/v1`, `jacobian_9D/derived_metrics/v1`, `finite_time_response/v1/{primary,stratified_9d}`, `transition_residuals/v1/{primary,stratified_9d}`, and `transition_residual_covariance_proxy/v1/{primary,stratified_9d}`,
- gated `stochastic_reachability/v1/{primary,stratified_9d}` surfaces driven only by an admissible one-step innovation covariance,
- separate dynamical-family surfaces under `dynamical_families/{diffusion,destination,resilience}/v1` when their configuration and qualification gates are satisfied,
- optional supporting groups such as `extensions`, `regions`, `regional_mnps`, and `manifest_json`.

This implemented contract is deliberately narrower than the theoretical compendium. Many paths are opt-in or gated, and a missing surface means unavailable rather than measured zero. In particular, stochastic reachability requires an admissible one-step transition covariance, committor requires declared destinations and estimator qualification, and FAR requires a finite perturbation protocol. All downstream inferences inherit raw-measurement, preprocessing, feature, chart, and estimator validity limits.

Family groups may also carry explicit `computation_status`, `measurement_validity`, and `claim_status` certificates. A computed value is not thereby a valid regime inference, and a valid measurement is not a biological claim. The operational contract does not promote these surfaces to an `ndt_licensed` result.

4. Cytoskeletal-Dendritic Accessibility Bridge

4.1. Role in the Layer Stack

The cytoskeletal-dendritic accessibility model adds a microstructural layer beneath MNPS without identifying dendrites with the noetic manifold itself. The updated Slow Branch article makes the bridge simulator-first: a slow, phenomenological branch variable M_b is written by eligibility, replay, and consolidation support, and is tested through memory-linking, perturbation, and rescue signatures [10]. The restricted claim is that this is an executable model-discrimination hypothesis, not an established biological foundation or an additional premise required for macro-level MNPS or local-dynamics inference. The current NDT evidence rests on the measurement stack, chart-level local dynamics, and bounded empirical accessibility analyses. Earlier C/TI/P summaries remain an optional reporting layer.

The layered dependency is:

- **Micro**: branch activity, fast access, eligibility traces, replay/consolidation support, and slow structural accessibility.
- **Meso**: a lower-dimensional accessibility geometry summarizing how branch support is distributed, concentrated, or canalised.
- **Chart**: provisional memory/accessibility coordinates, potentially 3D or stratified 9D, that can be analyzed with MNPS-style tools.
- **Macro**: local flow, MNJ structure, predictive-spread geometry, and optional regime reporting on the exported chart.

4.2. Branch-resolved simulator state

Let b index dendritic branches and μ, ν index memory traces. The updated simulator separates fast branch access from slow structural accessibility:

$$Z_t = \left\{ x_{b(t)}, A_b^{f(t)}, A_b^{s(t)}, E_{b(t)}, P_{b(t)}, M_{b(t)}, R_{\mu(t)}, L_{\mu\nu}(t) \right\}_{b,\mu,\nu} \tag{30}$$

Here $x_{b(t)}$ is fast branch activity, $A_b^{f(t)}$ is fast access, $A_b^{s(t)}$ is slow access derived from $M_{b(t)}$, $E_{b(t)}$ is eligibility, $P_{b(t)}$ is replay- and consolidation support, and $M_{b(t)}$ is the slow structural accessibility variable. $R_{\mu(t)}$ is trace-level recall support and $L_{\mu\nu}(t)$ is a pairwise linking output. None of these variables is claimed to be a direct molecular memory code.

The effective branch access factor is

$$A_{b(t)} = A_b^{f(t)} A_b^{s(t)} \tag{31}$$

This factorization is a gain-like simulator abstraction: fast cue and context can open a branch momentarily, while M_b biases how readily that opening is stabilized or reused.

At the fast timescale, branch activity follows

$$\tau_x \dot{x}_b = -x_b + A_{b(t)} \left[I_{b(t)} + R_{b(t)} \right] \tag{32}$$

where $I_{b(t)}$ is cue input and $R_{b(t)}$ is replay or recurrent recruitment. Eligibility and consolidation support follow

$$\tau_E \dot{E}_b = -E_b + \varphi(x_b, I_b) \tag{33}$$

and

$$\tau_P \dot{P}_b = -P_b + \rho_{\text{replay}} r_{b(t)} + \rho_\nu \nu(t) \tag{34}$$

respectively. Slow structural writing is then

$$\dot{M}_b = \eta E_{b(t)} P_{b(t)} W(t) (M_{\max} - M_b) - \lambda_M M_b + \varepsilon_{b(t)} \tag{35}$$

The bounded term expresses finite local capacity, $W(t)$ is the replay or writing window, λ_M captures structural decay, and $\varepsilon_{b(t)}$ is structural noise. The update is phenomenological: it is an executable model-discrimination rule, not a complete biochemical law.

For an individual trace and a pair of traces, the exported outputs are

$$R_{\mu(t)} = \sum_b a_{\mu b} A_{b(t)} x_{b(t)} \tag{36}$$

and

$$L_{\mu\nu}(t) = \sum_b a_{\mu b} a_{\nu b} M_{b(t)} \tag{37}$$

where $a_{\mu b}$ describes the allocation of trace μ to branch b. The simulator's joint profile is overlap-branch writing, linking gain, perturbation-sensitive loss, targeted rescue, and topology-dependent specificity [10]. It does not establish that dendrites or cytoskeletal structures directly store memory content.

4.3. Export to Mesoscopic Accessibility Geometry

The bridge to NDT begins by exporting the branch-resolved microstate to a mesoscopic memory/accessibility chart:

$$\Psi : Z_t \to x_t^{\mathrm{mem}} \tag{38}$$

Equation 38 is the explicit micro-to-meso interface. On that exported chart, one can define a local flow and Jacobian:

$$\dot{x}_t^{\mathrm{mem}} = f_{\mathrm{mem}(x_t^{\mathrm{mem}}, t)} + \varepsilon_t, \quad J_t^{\mathrm{mem}} = \frac{\partial f_{\mathrm{mem}}}{\partial x}(x_t^{\mathrm{mem}}, t) \tag{39}$$

The local flow in 39 can then be paired with the same short-horizon reachability recursion used for the measurement chart:

$$\Sigma_t^{(h+1)} = A_t \Sigma_t^{(h)} A_t^\top + W_{Q,t} \tag{40}$$

The important boundary condition is that $\Psi(Z_t)$ is an exported chart, not a claim that the branch microstate literally is the noetic manifold. Structural accessibility is hypothesized to shape the mesoscopic geometry that later appears as MNPS, MNJ, or reachability structure, while the macro measurement apparatus remains separable from any one molecular implementation. Until branch-resolved perturbation data directly link changes in Z_t or M_b to MNPS/MNJ/reachability shifts, this micro-to-NDT coupling remains speculative.

4.4. Coupling to Macro Metrics and Potentials

Let P be a projection from branch-level accessibility geometry into the regional or global MNPS chart. As a hypothesis-level extension, structural accessibility can enter macro geometry through effective mobility, diffusion, and potential terms:

$$M_{\mathrm{MNPS}(X,Z)} \approx P^\top M_{\mathrm{access}(\Psi(Z))} P, \quad a(X,t,Z) = a_0(X,t) + a_{\mathrm{access}(X,t,\Psi(Z))} \tag{41}$$

Equation 41 is not an identified biological law; it specifies where a future branch-resolved perturbation experiment could test the bridge.

In regional stratified notation, with regions r, s, chart coordinates α, β, and dendritic indices a, b:

$$M_{\mathrm{block}(r,s,\alpha,\beta;X,Z)} \approx \sum_a \sum_b P^r_{(a,\alpha)} {}_{\mathrm{access}(\Psi(Z))}M_{(a,b)} P^s_{(b,\beta)} \tag{42}$$

The corresponding drift split is

$$b(\xi, t \mid c, Z) = -M(\xi, t \mid c, Z) \nabla V(\xi, t \mid c, Z) + u_{\mathrm{nc}(\xi, t \mid c, Z)} \tag{43}$$

and the block-wise chart Jacobian is obtained by differentiating this full drift, not by assuming a gradient-only flow:

$$\hat{J}_{x,\mathrm{block}}(r, s, \alpha, \beta) = \frac{\partial \hat{b}_{x,r,\alpha}}{\partial x_{s,\beta}} \tag{44}$$

Equations 42 and 44 lift the same hypothesis to regional family blocks. This is the compendium-level bridge from micro to macro: slow branch-level accessibility may alter effective mobility, diffusion orientation, potential curvature, or transition gates, and those changes may then appear as altered MNJ blocks or predictive-spread geometry. The biological connection remains speculative until branch-resolved perturbation data identify it.

5. Meta–Noetic Jacobian (MNJ)

5.1. Definition

Empirical local flow is estimated on a declared measurement chart. Let $\hat{b}_x$ denote a chart-level drift estimate. The empirical Jacobian is

$$\hat{J}_{x(x,t)} = \frac{\partial \hat{b}_{x(x,t)}}{\partial x} \tag{45}$$

The corresponding latent derivative is $\nabla_\xi b(\xi, t)$. The term MNJ is reserved for a Jacobian estimated in an MNPS chart; an unadorned empirical J in this section means $\hat{J}_x$ unless a latent argument is explicit. Equation 45 is the chart-level derivative used throughout this section: it asks how nearby perturbations change under the local chart flow. MNJ does not by itself describe state location, stochastic spread, destination probability, or finite-amplitude resilience [[3]].

$$\hat{J}_{ij}, x(x,t) = \frac{\partial \hat{b}_{x,i}(x,t)}{\partial x_j} \tag{46}$$

For the base 3D chart,

$$\hat{J}_{x(x,t)} = \left(\left(\frac{\partial \hat{b}_{x,m}}{\partial m}, \frac{\partial \hat{b}_{x,m}}{\partial d}, \frac{\partial \hat{b}_{x,m}}{\partial e} \right) \left(\frac{\partial \hat{b}_{x,d}}{\partial m}, \frac{\partial \hat{b}_{x,d}}{\partial d}, \frac{\partial \hat{b}_{x,d}}{\partial e} \right) \left(\frac{\partial \hat{b}_{x,e}}{\partial m}, \frac{\partial \hat{b}_{x,e}}{\partial d}, \frac{\partial \hat{b}_{x,e}}{\partial e} \right) \right) \tag{47}$$

Equation 47 is the explicit 3D expansion of the component definition 46. The symmetric and antisymmetric parts are

$$S_x = \frac{1}{2}\left(\hat{J}_x + \hat{J}_x^\top\right), \quad \Omega_x = \frac{1}{2}\left(\hat{J}_x - \hat{J}_x^\top\right) \tag{48}$$

where S_x captures symmetric expansion and shear, while Ω_x captures an antisymmetric rotational component. Thus 48 decomposes the same chart Jacobian; it does not introduce a second local flow. Neither component should be interpreted as a unique biological mechanism.

5.2. Modal growth and reactivity families

The current local-dynamics vocabulary keeps modal stability and instantaneous reactivity separate:

$$\alpha(J) = \max_i \Re\left[\lambda_{i(J)}\right] \tag{49}$$

$$\omega(J) = \lambda_{\max\left(\frac{J+J^\top}{2}\right)} \tag{50}$$

$$\Delta_{\mathrm{react}(J)} = \omega(J) - \alpha(J) \tag{51}$$

Equations 49 and 50 define the two summaries, and 51 records their difference. The spectral abscissa α is the dominant modal growth or decay rate. The numerical abscissa ω is the maximal instantaneous growth rate. Thus $\alpha < 0$ and $\omega > 0$ can coexist: all eigenmodes decay while some perturbations initially grow. The reactivity gap is a derived, non-normality- sensitive summary of this separation, not a complete non-normality measure. The trace, spectral radius, Frobenius norm, rotation norm, and Henrici departure are additional summaries, not substitutes for α or ω.

Finite-time amplification is a separate family. A peak gain $G_{\mathrm{peak}(T)}$ is computed from a local propagator over a declared horizon and does not follow from an instantaneous eigenvalue summary. Truth-known simulator recovery of α, ω, and Δ_{react} under restricted linear systems is a methodological qualification; it is not empirical chart or biological validation [[4]].

5.3. Local Linear Estimation

The empirical MNJ estimator is based on a local linear flow model. A neighbourhood-wise fit over a declared chart and short analysis window takes the form [[3]; [11]; [12]]

$$\dot{x} \approx A(x - x_{\mathrm{ref}}) + b_x \tag{52}$$

with

$$\hat{J}_x = A \tag{53}$$

The fitted matrix in 52 is identified with the Jacobian estimate by 53. In the NMD contract this is a chart-level estimate, typically regularized over local neighbours or short super-windows. The serialized `mnps_3d_dot` is an input or derivative surface, not by itself an independently identified latent drift. A temporal difference of J is likewise not automatically the latent derivative $\dot{J}$.

For discrete-time maps one also uses the continuous/discrete relation

$$J_d = e^{J_c \Delta t}, \quad J_c \approx \frac{J_d - I}{\Delta t} \tag{54}$$

Equation 54 relates discrete and continuous local operators only in the small-Δt regime, subject to the declared sampling and model contract.

5.4. Conditional relation to potential curvature

If the drift split in 9 is declared and M is locally constant, the potential component contributes a Hessian-shaped term. In the general state-dependent case, differentiating

$$b = -M\nabla V + u_{\mathrm{nc}} \tag{55}$$

gives schematically

$$J_b = -M\nabla^2 V - (\partial M)\nabla V + \partial u_{\mathrm{nc}} \tag{56}$$

where $(\partial M)\nabla V$ denotes the collection of mobility-variation terms. Therefore

$$J \approx -M\nabla^2 V \tag{57}$$

is only a conditional approximation. Chart Jacobian asymmetry does not by itself identify u_{nc} because state-dependent mobility can also produce it.

5.5. 9D Jacobians and Family Blocks

On the stratified chart the Jacobian becomes a 9×9 local linearization. The m, d, and e coordinate families can be grouped into a 3×3 family-block matrix, as shown in 58:

$$J^{(9)} = \begin{pmatrix} J_{\mathrm{m,m}} & J_{\mathrm{m,d}} & J_{\mathrm{m,e}} \\ J_{\mathrm{d,m}} & J_{\mathrm{d,d}} & J_{\mathrm{d,e}} \\ J_{\mathrm{e,m}} & J_{\mathrm{e,d}} & J_{\mathrm{e,e}} \end{pmatrix} \tag{58}$$

where each $J_{\mathrm{out,in}} \in \mathbb{R}^{3\times 3}$ describes how in-family variation drives out-family rates of change. This block view is useful when global 3D coordinates appear unchanged but mechanism-specific subcoordinates redistribute. The blocks are chart-level summaries and do not establish anatomical coupling.

6. Geometry of Possible Futures and Local Capacity

6.1. Local Stochastic Linearization

Stochastic short-horizon reachability formalizes how probability mass spreads around a state over a finite horizon, rather than describing the state's current location alone. It is a covariance-based approximation to predictive spread under the chart-conditional law 8, not classical control-theoretic reachability and not a structural-connectome Gramian. A local discrete-time model is [[13]; [14]].

$$x_{t+1} = A_t x_t + b_t + \varepsilon_t, \quad \varepsilon_t \sim \left(0, W_{Q,t}\right) \tag{59}$$

In 59, A_t and b_t are estimated locally in chart-space neighbourhoods and $W_{Q,t}$ is an admissible one-step transition-innovation covariance. MNPS asks where the system is, MNJ asks how the local chart flow transforms nearby states, and this family asks how finite-horizon predictive mass spreads. Condition mixtures or poorly qualified residuals can inflate W_Q without establishing richer controllable geometry.

6.2. Covariance Recursion and Reachable Sets

The reachability covariance is initialized and propagated by

$$\Sigma_t^{(0)} = 0, \quad \Sigma_t^{(h+1)} = A_t \Sigma_t^{(h)} A_t^{\top} + W_{Q,t} \tag{60}$$

Equation 60 propagates the covariance from the one-step model 59. The corresponding predictive-spread ellipsoid at horizon h is

$$\mathcal{E}_t^{(h)}(\alpha) = \left\{ z \mid z^{\top} \left(\Sigma_t^{(h)}\right)^{-1} z \le \alpha \right\} \tag{61}$$

These objects may be reported at fixed horizons or aggregated across several horizons into a tube summary. They describe second-moment spread under a local linearization, not the support of every admissible control. Horizon choice, neighbourhood leakage, and innovation-residual quality are validity conditions. An NMD reachability surface is therefore gated and is not implied by the presence of a chart or Jacobian.

6.3. Primary Cone Metrics

Let $\lambda_1 \geq ... \geq \lambda_d$ denote the eigenvalues of $\Sigma_t^{(h)}$. The primary metrics are:

$$\mathrm{Vol}_t^{(h)} = \frac{1}{2}\log\Big(\det\Big(\Sigma_t^{(h)}\Big)\Big) \tag{62}$$

$$\kappa_t^{(h)} = \frac{\lambda_1}{\lambda_d + \varepsilon} \tag{63}$$

$$d_{\text{eff}}, t^{(h)} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2} \tag{64}$$

For rotation, the cone article recommends a robust discrete proxy. If $A_t = W\Sigma V^T$ is the SVD of A_t, define $U_t = WV^T$ and

$$\rho_t^{\mathrm{A}} = \left\| \frac{U_t - U_t^T}{2} \right\|_{\mathrm{F}} \tag{65}$$

When a stable continuous-time MNJ exists, the optional Jacobian-based rotation summary is

$$\rho_t = \left\| \frac{J_t - J_t^T}{2} \right\|_{\mathrm{F}} \tag{66}$$

The minimal persistence formalization is expressed on a tube metric $V_t = \mathrm{tube_log_det}(t)$:

$$\tau_{\mathrm{rec}(t)} = \inf_{\Delta t > 0} \{\Delta t : V_{t+\Delta t} \geq V_{\mathrm{rec}}\} \tag{67}$$

with thresholds defined relative to a baseline quantile of the tube-volume series. Taken together, these metrics decompose capacity into size, directional concentration, effective rank, rotational structure, and recovery dynamics [[13]; [14]; [15]; [16]; [17]].

6.4. Secondary Cone Metrics

The reachability manuscript also defines two useful secondary summaries:

$$p_i = \frac{\lambda_i + \varepsilon}{\sum_j \lambda_j + d\varepsilon}, \quad H_{\mathrm{dir}} = -\sum_i p_i \log(p_i), \quad \tilde{H}_{\mathrm{dir}} = \frac{H_{\mathrm{dir}}}{\log(d) + \varepsilon} \tag{68}$$

$$r_{\mathrm{vol}(t)} = \exp(\mathrm{tube_log_det}(t) - \mathrm{tube_log_det}(t-1)) \tag{69}$$

Directional entropy summarizes how evenly reachability mass is distributed across directions; the transport ratio summarizes local expansion or shrinkage between adjacent windows.

6.5. Interpretation Discipline

The cone layer is explicitly capacity-focused. It is not a direct predictor of where the system will go, nor is it identical to occupancy, entropy, or speed. The most important synthesis rule from the source papers is:

- occupancy and reachability can dissociate,
- traversability and reachability are related but not identical,
- heteroskedastic innovation covariance $W_{Q,t}$ can inflate cones without implying richer controllable geometry,
- horizon dependence and neighborhood leakage must be treated as validity conditions, not as afterthoughts.

6.6. Committor geometry

For competing destinations A and B, the committor is

$$q_{AB}(x) = P_{x(\tau_B < \tau_A)} \tag{70}$$

It is a path-dependent hitting probability and therefore destination geometry, not local expansion, diffusion volume, or traversability. Empirical use requires explicit disjoint destinations, independent segments reaching both regimes, a reaction-coordinate contract, and estimator qualification. Stage labels alone are not committor truth. Finite-sample two-dimensional recovery is not a currently validated NMD export; `mndm.committor.v1` remains a gated schema or adapter rather than a claim of committor identification [[18]; [19]; [20]].

6.7. Attractors and finite-amplitude resilience

Attractors and basins describe longer-time organization: locations, boundaries, metastability, escape, and preferred corridors. A visual cluster is not by itself evidence for an attractor. Finite-amplitude resilience (FAR) asks how large a declared perturbation the system can tolerate before leaving its dynamical regime. It requires

an observed perturbation protocol and amplitude family. Local Jacobian stability, finite-time tangent gain, and spontaneous trajectory variation cannot substitute for that protocol.

7. Optional C/TI/P Regime Reporting

This section preserves an earlier regime-reporting layer in the article family. It is not one of the core local-dynamics families in the current NDT article and it must not be collapsed into a consciousness scalar.

The revised Churchland synthesis adds a regime-classification layer above MNPS, MNJ, traversability, and reachability. The point is to avoid replacing one scalar consciousness marker with another. Entropy, occupancy, speed, and local capacity can all be informative, but none is sufficient on its own. The relevant object is a three-axis regime geometry:

- **Capacity collapse (C)**: reduced local accessibility, traversal speed, reachability volume, effective dimensionality, rotational structure, or adaptive availability.
- **Transport-innovation balance (TI)**: whether low capacity remains innovation-dominated or becomes captured by local transport dynamics.
- **Pathological lock-in (P)**: whether the regime is rigid, canalised, and asymmetrically deformative in a pathology-like way rather than simply physiologically constrained.

In compact notation, define a regime descriptor

$$R_t^{\mathrm{CTIP}} = (C_t, \mathrm{TI}_t, P_t) \tag{71}$$

Equation 71 defines the reporting tuple; each component is computed from families of reported summaries rather than from a single privileged scalar. A minimal operational reading is:

$$C_t = \mathcal{C}(\mathrm{speed}, \mathrm{tube_log_det}, d_{\mathrm{eff}}, \rho), \quad \mathrm{TI}_t = \mathcal{TI}(\text{Q-ratio}, \mathrm{CaptureGate}, \mathrm{MDR}), \quad P_t = \mathcal{P}(\mathrm{ACI}, \mathrm{CFCA}, \mathrm{asymmetry}) \tag{72}$$

Equation 72 gives one compact decomposition of the three components in 71. The exact functions $\mathcal{C}$, $\mathcal{TI}$, and $\mathcal{P}$ are contract-level reporting maps, not universal constants. Their role is to force every reduced state claim to say whether it is reporting capacity loss, transport capture, pathological deformation, or some mixture of the three.

7.1. Core Regime Dissociation

The present article family uses sleep, propofol, and epilepsy as the main stress test for the C/TI/P layer:

Regime	C: capacity	TI: transport/innovation	P: lock-in
Deep NREM sleep	Low capacity relative to wake/REM	Comparatively innovation-dominated in current analyses	Weak or non-significant pathological lock-in
Propofol	Capacity separation, strongest for awake vs sedated	No clean corrected transport-capture profile in present dose range	No clean corrected pathological-lock profile
Ictal epilepsy	Strong collapse in speed, cone volume, and effective dimensionality	Transport capture via lower `Q-ratio_H4` and higher `CaptureGate`	Reachability-canalisation and cross-family coupling asymmetry increase

Table 1: Core regime dissociation across capacity, transport/innovation, and pathological lock-in.

The rows in Table 1 are a reporting matrix, not a new dynamical family.

The key conclusion is not that capacity is irrelevant. It is that capacity collapse is underdetermined. Deep sleep, propofol sedation, and ictal epilepsy can all reduce local availability, but they do not imply the same biological or consciousness-relevant state unless TI and P are also reported.

This distinction also resolves a potential anisotropy ambiguity. `ACI` in the C/TI/P layer denotes reachability-canalisation: whether short-horizon reachable futures concentrate into a narrow subset of directions. This is distinct from dispersion of local Jacobian eigenvalues. A seizure regime can therefore show uniformly contractive local Jacobian spectra while still exhibiting high reachability-canalisation among the few futures that remain locally admissible.

8. Traversability, Extensions, and Modality Boundaries

8.1. Traversability Index

The Churchland paper packages speed and anisotropy into an operational TraversabilityIndex:

$$T = v \cdot (1 - A_{\text{norm}}) \tag{73}$$

Equation 73 combines speed and normalized anisotropy, with cohort-relative normalization

$$A_{\text{norm}} = \frac{A - A_{\min}}{A_{\max} - A_{\min}} \tag{74}$$

as defined in 74. Low traversability can therefore arise from slow motion, strong canalization, or both. For this reason the compendium treats T as a compact diagnostic, not as a substitute for explicit Jacobian or reachability geometry. Its real value in the article family is comparative: it gives a one-line summary that can be reported alongside richer dynamical objects without pretending to replace them.

8.2. Jump-Diffusion Extension

Several manuscripts sketch an enriched controlled dynamics with rare discontinuous events:

$$dX_t = b(X_t, t \mid c_t)dt + \Sigma(X_t, t \mid c_t)dW_t + \int_{\mathbb{R}^d} \kappa(X_{t^-}, z)\tilde{N}(dt, dz) \tag{75}$$

Equation 75 extends the backbone in 4 with a jump measure. This extension is kept secondary: it can model abrupt regime shifts, spikes, or other rare transitions without replacing the controlled-diffusion backbone.

8.3. Modality-Specific Identifiability

The theoretical framework is modality-general, but its empirical second-order layer is not equally identifiable across modalities:

- EEG and iEEG can support first-order and second-order estimation when the chart, sampling, residual quality, and neighbourhood contract are adequate; availability does not establish operator validity.
- fMRI generally resolves first-order geometry more reliably than second-order flow. MNJ and predictive-spread analyses in BOLD are boundary-condition probes that demand stronger validity checks.

This asymmetry is a core cross-paper discipline: weak or near-null second-order effects in fMRI should not be overinterpreted as either strong confirmation or strong falsification of the framework.

9. Compact Synthesis

The formal dependency structure of the article family can be summarized as:

1. A latent stochastic process evolves on $\mathcal{M}$ as a controlled diffusion (Eqs. 4–8).
2. Measurements Y_t are mapped into a base or stratified chart (Eqs. 3–29).
3. Local chart derivatives define MNJ and its modal or reactivity summaries (Eqs. 45–58). When the cytodendritic bridge is active, branch-level accessibility can enter through exported metric/potential terms (Eqs. 30–44).
4. Local diffusion and one-step innovations define gated covariance propagation and short-horizon predictive-spread geometry (Eqs. 6–60).
5. Committor, attractor, and FAR families address destination, long-time, and finite-amplitude questions under separate protocols.
6. Candidate controls act on drift, diffusion, mobility, or transition gates; RVC is only the isotropic variance-control special case.
7. Optional C/TI/P and traversability summaries report combinations of these layers without replacing them.

Within this unified view, the family of papers converges on one central claim: conscious-state analysis is not adequately described by entropy, coherence, or position alone. The mathematically relevant objects are the joint structure of state coordinates, local laws, diffusion geometry, chart derivatives, finite-horizon predictive spread, destination geometry, and conditional access control. Put slightly differently, the current article becomes more informative as it moves from state location to local law, to non-equivalent future-law families, and then to the controls that can reshape them [[1]; [2]; [3]; [4]].

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13. Data and Code Availability

Reference implementation for the measurement layer: `https://github.com/NoeticDiffusion/NeuralManifoldDynamics`

Related Noetic Diffusion repositories and manuscript sources: `https://github.com/NoeticDiffusion/`

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